FINAL YEAR PROJECT PROPOSAL

# AUTOMATED CODE VULNERABILITY ASSESSMENT TOOL USING LARGE LANGUAGE MODELS

INU


**Submitted By:**

**Jawad Ahmed** (21990)

**Sami Ullah** (21994)


**Session 2022-2026**


**Supervised By:**

***Engr. Mubashir Zainoor***


A project proposal submitted to The IQRA National University, Peshawar in partial fulfillment of the requirements for the degree of

**BACHELOR OF SCIENCE (BS) IN COMPUTER SCIENCE**


SCHOOL OF COMPUTER SCIENCE

**IQRA NATIONAL UNIVERSITY, PESHAWAR-PAKISTAN**


December, 2025

## Table of Contents


1. Introduction & Problem Statement 1

2. Objectives of Project 1

3. Methodology & System Architecture 2

4. Proposed Time Schedule (Gantt Chart) 3

5. Budget Description & Technical Stack 3

6. Undertaking & Signatures 4


## List of Acronyms

| Acronym | Description |
|---|---|
| INU | IQRA National University |
| AI | Artificial Intelligence |
| NLP | Natural Language Processing |
| LLM | Large Language Model |
| API | Application Programming Interface |
| UI / UX | User Interface / User Experience |
| BLE / IoT | Bluetooth Low Energy / Internet of Things |

## 1. Introduction & Problem Statement

The project titled **"Automated Code Vulnerability Assessment Tool Using Large Language Models"** addresses critical architectural paradigms within the modern computer science landscape. As information engineering systems shift toward distributed, agile environments, specialized automated monitoring and optimization models become absolutely paramount to unlock next-generation performance.

**Problem Statement:** Fast software development deployment cycles often overlook fundamental security check protocols, pushing vulnerable code structures to live production. Traditional static code evaluation setups (SAST) return massive volumes of distracting false positives and fail to understand structural logic flows across interconnected dependencies. Without an automated, reliable system structure to

streamline these processes, operational risks escalate dramatically. This system solves this missing gap by developing an optimized computational pipeline that acts as a robust mechanism for resolving these complex challenges.

## 2. Objectives of Project

The primary structural milestones mapped for this software development cycle are:

- Construct an automated Abstract Syntax Tree (AST) parsing infrastructure mapping codebase repository layouts cleanly.
- Fine-tune an offline, instruction-tuned LLM optimized specifically to detect edge-case logic and structural memory leak threats.
- Incorporate automated secure patch generation mechanisms detailing refactoring solutions natively within development pipelines.

## 3. Methodology & System Architecture

The system adheres to a structured, modular pipeline layout designed using a decoupled Client-Server model. This ensures that heavy backend model inferencing jobs do not impair the frontend interface application execution parameters.

### Core Modules:

1. **Authentication & Profile Integration Module:** Handles user record enrollment and data secure caching sequences smoothly.
2. **Core Execution and Inference Pipeline:** Manages real-time data ingestion, transformation operations, and algorithmic processing nodes.
3. **Analytics & Reporting Layout Engine:** Aggregates calculation products, compiling structured system reports safely.

*[System Architecture Design: React Native Client Interface ←→ Secure Gateway APIs ←→ Python/ Node Compute Processing Engines ←→ Scalable Vector/Relational Databases]*

## 4. Proposed Time Schedule

| Phase | Activity Description | Duration (2025-2026) |
|---|---|---|
| Phase 1 | Literature Survey & Component Finalization | December |
| Phase 2 | Architecture Modeling & Engineering Checks | January - February |
| Phase 3 | Core Engine Development & Coding Cycle | March - April |
| Phase 4 | System Verification Testing & Report Synthesis | May - June |

### Visual Timeline (Gantt Chart Representation)

| Activity | Dec | Jan | Feb | Mar | Apr | May |
|---|---|---|---|---|---|---|
| Literature Gathering | ■ | | | | | |
| Scheme Design Modeling | | ■ | ■ | | | |
| System Core Coding | | | | ■ | ■ | |
| Verification & Writing | | | | | | ■ |

## 5. Budget Description & Technical Stack

The project leverages open-source libraries and public cloud deployment tiers, achieving maximum software efficiency with minimal capital expenditures. All computation and model runs are hosted on standard developer hardware platforms already in position.

**Proposed Technical Core Stack:** React Native Dev Panel, Node.js Core, HuggingFace Transformers, Tree-Sitter Parser, LangChain, Docker Sandboxing.

## 6. Undertaking & Project Declarations

We hereby declare that this technical development proposal is our genuine work executed under supervision at IQRA National University. No portion of this engineering research layout has been submitted previously for academic credit at any alternative educational board.

Student 1 Signature

Student 2 Signature

## 7. Institutional Review Panel Logs

**Supervisor Evaluation Section:**

_____

Supervisor Signature Date: ____/____/_____

**FYP Department Coordinator Workspace:**

_____

Coordinator Signature Status: APPROVED / HOLD